Quantum Teleportation: A Portal to the Quantum Age

Jolana Kampfová Abdulaziz Sobirov Dafne Unsal Nuchi

December 17, 2024

1 Introduction

During the Solvay congress of 1927, the most brilliant phycisists of the moment got together to discuss the latest advances in the field. Quantum physics, which was in its early stages of development, was naturally one of the main topics of discussion. During that congress, Albert Einstein, father of quantum physics and also one of its biggest critics, and Niels Bohr, a clear defender of quantum physics and the Copenhagen interpretation, got into several discussions about one specific phenomenon that this theory predicted – quantum entanglement [4].

This phenomenon seemingly allowed for the intrinsic linkage of two particles' states, making it impossible to describe them individually – and, by making a measurement on one of them and collapsing the wave function, the other's state could be inferred. This seems extremely counterintuitive for us and doesn't have anything comparable in the scale in which we live, which caused the rejection of the theory by many physicists, including Einstein. In fact, Einstein's dislike for quantum physics eventually culminated in him writing a paper with Boris Podolsky and Nathan Rosen that showed a paradox within quantum physics (the famous EPR paradox) [2]. This paradox poses the situation of two observers, each of which has one member of a pair of entangled particles. If both are very far away, then, when one measures their particle, the other particle's state will also be known to them. Einstein argued that this broke Special Relativity by allowing information to travel faster than light, and therefore this was not what was happening. He advocated for the existence of "hidden variables", which were hiding the information that quantum physics modeled as undetermined until measured [5]. The paper also argued that this situation would allow to measure one of the particles' observables (for example, position) and therefore know the other's position too, without ever disturbing the second one. This would lead to the breaking of Heisenberg's uncertainty principle, since we would be able to measure the particle's momentum and therefore know its position and momentum simultaneously [1].

This seemed like a huge blow for quantum physics, but the paradox was eventually solved. Einstein's argument of information travelling faster than light was not accurate, since the second observer didn't know what his particle's state was, and therefore no information was transmitted between both observers and nothing was travelling faster than light. His argument that entanglement broke the uncertainty principle was also found to be false [1], and to top it all off, a physicist of the name John Bell was able to define a series of inequalities (known as the Bell inequalities) that could be measured experimentally, and could determine which model was correct – quantum physics or hidden variables [1]. As the experiments were conducted, quantum mechanics emerged victorious, and Einstein was proven wrong [4] [5].

Nowadays, quantum entanglement is accepted and even used by scientists on a regular basis. One of its most interesting applications is in computation, with the creation of concepts like quantum internet or quantum cryptography. Both of these use a very particular effect of quantum entanglement, known as quantum teleportation. This strange phenomenon lets us transport information in counterintuitive ways, opening new possibilities for communication. This bizarre phenomenon that angered Einstein so much is nowadays extremely relevant, but how does it work, exactly? And, if we said that quantum entanglement doesn't send any information between two observers, then how can we use it to do just that? These are the questions that we will answer in our project, and we will then connect back to the new technologies we have mentioned, which use quantum teleportation to their advantage.

2 How does quantum teleportation work?

Quantum teleportation is complex, so we will be visualizing all of our cases with experiments, with Alice and Bob as our experimentalists. Let's say that Alice has acquired a single qubit (the basic information unit in quantum computing). We can represent this with the qubit's state vector:

$$|\Psi\rangle = \alpha|1\rangle + \beta|0\rangle$$

This qubit, as we can see, is in a superposition between states 1 and 0, where α and β are complex numbers. According to the born rule, $|\alpha|^2$ is the probability of a measurement giving us state 1, and $|\beta|^2$ is the probability of getting 0 when we measure. Therefore,

$$|\alpha|^2 + |\beta|^2 = 1$$

. We can also represent this in matrix notation, by assigning the state $|1\rangle$ the matrix $\begin{pmatrix}1\\0\end{pmatrix}$ and the state $|0\rangle$ the matrix $\begin{pmatrix}0\\1\end{pmatrix}$:

$$|\Psi\rangle = \alpha|1\rangle + \beta|0\rangle \rightarrow \begin{pmatrix}\alpha\\\beta\end{pmatrix}$$

Let us now say that Bob also has a qubit (we will label Alice's qubit with an A and Bob's qubit with a B to set up our example). If Alice's qubit is in state 1, and Bob's is in state 0, we can represent this system with a composite state vector:

$$|\psi_{AB}\rangle = |1_A\rangle|0_B\rangle$$

As we can see, we represent this new state vector with the product between the two states of the qubits. If Alice were to measure her qubit in this state, it would result in her qubit collapsing in the state 1 with a probability of 100% (since we set up the qubit to be in this state). The state vector, in this case, would just become $|0_B\rangle$. If Bob does the same, then he would get the state 0 with 100% of probability. In this case, the state vector would collapse to $|1_A\rangle$. This shows us that, even though we wrote the two qubits into a same state vector, they are independent of each other, and can be collapsed independently without affecting the other. If our qubits weren't defined from the beginning, we wouldn't be able to know the state of one by collapsing the other.

Now, let us consider another case. Alice and Bob again have their qubits in the states 1A and 0B, but now their composite state vector looks like this:

$$|\psi^+_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|0_B\rangle + \frac{1}{\sqrt{2}}|0_A\rangle|1_B\rangle$$

If we do the same experiment now, we will see that things have changed. Let us say Alice measures her qubit – which, again, because of the setup, will give us state 1. Once measured, the qubit will collapse into state 1, and therefore it will forcibly collapse into the first term of the sum. But, because the second term disappears, we suddenly know that the qubit B has to be in state 0, since the term $|1_B\rangle$ disappears from the equation. In this case, because $|1_A\rangle$ and $|0_B\rangle$ are both one after the measurement of one of them, the final state vector just turns out to be $\frac{1}{\sqrt{2}}$. We can see that, in this case, the measurement and collapse of one of the qubits gives us information about the other, and we can't separate them from each other in the formula. This is an example of two entangled qubits, whose states are intrinsically linked.

We can now construct 4 composite state vectors for the qubits A and B using the four basis state vectors that we have:

$$|1_A\rangle|1_B\rangle \rightarrow \begin{pmatrix}1\\0\\0\\0\end{pmatrix}$$

$$|1_A\rangle|0_B\rangle \rightarrow \begin{pmatrix}0\\1\\0\\0\end{pmatrix}$$

$$|0_A\rangle|1_B\rangle \rightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$|0_A\rangle|0_B\rangle \rightarrow \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

in order to acquire a basis for the overall system's Hilbert space in terms of the 4 basis state vectors $|1_A\rangle|1_B\rangle$, $|0_A\rangle|1_B\rangle$, $|1_A\rangle|0_B\rangle$, $|0_A\rangle|0_B\rangle$. So, let us rewrite the 2 state vectors that we saw earlier ($|\Psi_{AB}\rangle$ and $|\Psi^+_{AB}\rangle$) in matrix notation:

$$|\psi_{AB}\rangle = |1_A\rangle|0_B\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|\psi^+_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|0_B\rangle + \frac{1}{\sqrt{2}}|0_A\rangle|1_B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

This is just an alternative way to represent the state vectors which will come in handy later. Now, let us generalize the entangled state vector $|\psi^+_{AB}\rangle$ that we looked at into four similar entangled state vectors

$$|\psi^+_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|0_B\rangle + \frac{1}{\sqrt{2}}|0_A\rangle|1_B\rangle$$

$$|\psi^-_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|0_B\rangle - \frac{1}{\sqrt{2}}|0_A\rangle|1_B\rangle$$

$$|\phi^+_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|1_B\rangle + \frac{1}{\sqrt{2}}|0_A\rangle|0_B\rangle$$

$$|\phi^-_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|1_B\rangle - \frac{1}{\sqrt{2}}|0_A\rangle|0_B\rangle$$

If we rewrite these 4 state vectors using the matrix notation that we introduced earlier

$$|\psi^+_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|0_B\rangle + \frac{1}{\sqrt{2}}|0_A\rangle|1_B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

$$|\psi^-_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|0_B\rangle - \frac{1}{\sqrt{2}}|0_A\rangle|1_B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} - \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

$$|\phi^+_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|1_B\rangle + \frac{1}{\sqrt{2}}|0_A\rangle|0_B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$|\phi^-_{AB}\rangle = \frac{1}{\sqrt{2}}|1_A\rangle|1_B\rangle - \frac{1}{\sqrt{2}}|0_A\rangle|0_B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} - \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

we can see that there is an important property among the four of them: they're all orthonormal. Here are the explicit calculations to demonstrate it:

$$\langle\psi^+_{AB}|\psi^+_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & 1 & 1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\\0\end{pmatrix} = \frac{1}{2}+\frac{1}{2} = 1$$

$$\langle\psi^-_{AB}|\psi^-_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & 1 & -1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\\0\end{pmatrix} = \frac{1}{2}+\frac{1}{2} = 1$$

$$\langle\phi^+_{AB}|\phi^+_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 0 & 1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix} = \frac{1}{2}+\frac{1}{2} = 1$$

$$\langle\phi^-_{AB}|\phi^-_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 0 & -1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\-1\end{pmatrix} = \frac{1}{2}+\frac{1}{2} = 1$$

$$\langle\psi^+_{AB}|\psi^-_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & 1 & 1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\\0\end{pmatrix} = \frac{1}{2}-\frac{1}{2} = 0$$

$$\langle\psi^+_{AB}|\phi^+_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & 1 & 1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix} = 0$$

$$\langle\psi^+_{AB}|\phi^-_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & 1 & 1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\-1\end{pmatrix} = 0$$

$$\langle\psi^-_{AB}|\psi^+_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & 1 & -1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\\0\end{pmatrix} = \frac{1}{2}-\frac{1}{2} = 0$$

$$\langle\psi^-_{AB}|\phi^+_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & 1 & -1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix} = 0$$

$$\langle\psi^-_{AB}|\phi^-_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & 1 & -1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\-1\end{pmatrix} = 0$$

$$\langle\phi^+_{AB}|\psi^+_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 0 & 1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\\0\end{pmatrix} = 0$$

$$\langle\phi^+_{AB}|\psi^-_{AB}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 0 & 1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\\0\end{pmatrix} = 0$$

$$\langle \phi^+_{AB} | \phi^-_{AB} \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} = \frac{1}{2} - \frac{1}{2} = 0$$

$$\langle \phi^-_{AB} | \psi^+_{AB} \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$\langle \phi^-_{AB} | \psi^-_{AB} \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} = 0$$

$$\langle \phi^-_{AB} | \phi^+_{AB} \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{2} - \frac{1}{2} = 0$$

The orthonormality of the four state vectors (which are also known as Bell state vectors) will come in handy in our later explanations. With these four vectors, we can represent an arbitrary composite state vector that is expressed in terms of the four basis state vectors ($|1_A\rangle|1_B\rangle$, $|1_A\rangle|0_B\rangle$, ...) as a linear combination of those 4 entangled and orthonormal state vectors

$$|1_A\rangle|0_B\rangle = \frac{\sqrt{2}}{2}(|\psi^+_{AB}\rangle + |\psi^-_{AB}\rangle)$$

$$|0_A\rangle|1_B\rangle = \frac{\sqrt{2}}{2}(|\psi^+_{AB}\rangle - |\psi^-_{AB}\rangle)$$

$$|1_A\rangle|1_B\rangle = \frac{\sqrt{2}}{2}(|\phi^+_{AB}\rangle + |\phi^-_{AB}\rangle)$$

$$|0_A\rangle|0_B\rangle = \frac{\sqrt{2}}{2}(|\phi^+_{AB}\rangle - |\phi^-_{AB}\rangle)$$

Here is an example of an arbitrary composite state vector expressed in that form:

$$\begin{aligned} |\Psi_{AB}\rangle &= \alpha|1_A\rangle|1_B\rangle + \beta|1_A\rangle|0_B\rangle + \gamma|0_A\rangle|1_B\rangle + \delta|0_A\rangle|0_B\rangle \\ &= \frac{\sqrt{2}}{2}\alpha(|\phi^+_{AB}\rangle + |\phi^-_{AB}\rangle) + \frac{\sqrt{2}}{2}\beta(|\psi^+_{AB}\rangle + |\psi^-_{AB}\rangle) + \frac{\sqrt{2}}{2}\gamma(|\psi^+_{AB}\rangle - |\psi^-_{AB}\rangle) + \frac{\sqrt{2}}{2}\delta(|\phi^+_{AB}\rangle - |\phi^-_{AB}\rangle) \end{aligned}$$

As we can see, the original state vector is now expressed in terms of the 4 entangled and orthonormal state vectors rather than the 4 basis state vectors. This provides us with an alternative orthonormal basis to express the overall Hilbert space of the system, which will prove extremely useful to assume certain postulates in our later explanations.

Consider now a scenario where Alice and Bob prepare entangled pairs of qubits A and B with an entangled state vector $|\psi^+_{AB}\rangle$ and then move far away from each other in space. Furthermore, consider that Alice has a third qubit, A', which is expressed in an unknown quantum state vector:

$$|\Psi_{A'}\rangle = \alpha|1_{A'}\rangle + \beta|0_{A'}\rangle$$

Given this situation, the quantum state vector for the overall three qubit system $|\Psi_{A'AB}\rangle$ is:

$$|\Psi_{A'AB}\rangle = |\Psi_{A'}\rangle|\psi^+_{AB}\rangle = (\alpha|1_{A'}\rangle + \beta|0_{A'}\rangle)(\frac{1}{\sqrt{2}}|1_A\rangle|0_B\rangle + \frac{1}{\sqrt{2}}|0_A\rangle|1_B\rangle)$$

As we can see, there is entanglement between qubits A and B, but no entanglement with qubit A'. Now let us imagine that Alice makes a measurement in her qubit A', measuring for some observable

$\mathcal{O}$, that has four possible outcomes, each of which corresponds to one of the four Bell state vectors $|\psi^+_{A'A}\rangle$, $|\psi^-_{A'A}\rangle$, $|\phi^+_{A'A}\rangle$, and $|\phi^-_{A'A}\rangle$. Let us see what happens once she makes this measurement. Since we saw earlier that we can express a composite system as a linear combination of 4 state entangled and orthonormal state vectors, let us perform the same operation on this composite three-qubit system, only using the four Bell state vectors $|\psi^+_{A'A}\rangle$, $|\psi^-_{A'A}\rangle$, $|\phi^+_{A'A}\rangle$, and $|\phi^-_{A'A}\rangle$, which are the same as before, but with A' replacing A and A replacing B:

$$|1_{A'}\rangle|0_A\rangle = \frac{\sqrt{2}}{2}(|\psi^+_{A'A}\rangle + |\psi^-_{A'A}\rangle)$$

$$|0_{A'}\rangle|1_A\rangle = \frac{\sqrt{2}}{2}(|\psi^+_{A'A}\rangle - |\psi^-_{A'A}\rangle)$$

$$|1_{A'}\rangle|1_A\rangle = \frac{\sqrt{2}}{2}(|\phi^+_{A'A}\rangle + |\phi^-_{A'A}\rangle)$$

$$|0_{A'}\rangle|0_A\rangle = \frac{\sqrt{2}}{2}(|\phi^+_{A'A}\rangle - |\phi^-_{A'A}\rangle)$$

With this, we can write

$$\begin{aligned}
|\Psi_{A'AB}\rangle &= |\Psi_{A'}\rangle|\psi^+_{AB}\rangle = (\alpha|1_{A'}\rangle + \beta|0_{A'}\rangle)(\frac{1}{\sqrt{2}}|1_A\rangle|0_B\rangle + \frac{1}{\sqrt{2}}|0_A\rangle|1_B\rangle) \\
&= \frac{1}{\sqrt{2}}\alpha|1_{A'}\rangle|1_A\rangle|0_B\rangle + \frac{1}{\sqrt{2}}\alpha|1_{A'}\rangle|0_A\rangle|1_B\rangle + \frac{1}{\sqrt{2}}\beta|0_{A'}\rangle|1_A\rangle|0_B\rangle + \frac{1}{\sqrt{2}}\beta|0_{A'}\rangle|0_A\rangle|1_B\rangle \\
&= \frac{1}{\sqrt{2}}\frac{\sqrt{2}}{2}(|\phi^+_{A'A}\rangle + |\phi^-_{A'A}\rangle)|0_B\rangle + \frac{1}{\sqrt{2}}\frac{\sqrt{2}}{2}(|\psi^+_{A'A}\rangle + |\psi^-_{A'A}\rangle)|1_B\rangle \\
&+ \frac{1}{\sqrt{2}}\frac{\sqrt{2}}{2}(|\psi^+_{A'A}\rangle - |\psi^-_{A'A}\rangle) + \frac{1}{\sqrt{2}}\frac{\sqrt{2}}{2}(|\phi^+_{A'A}\rangle - |\phi^-_{A'A}\rangle) \\
&= \frac{1}{2}|\psi^+_{A'A}\rangle\alpha|1_B\rangle + \frac{1}{2}|\psi^+_{A'A}\rangle\beta|0_B\rangle + \frac{1}{2}|\psi^-_{A'A}\rangle\alpha|1_B\rangle - \frac{1}{2}|\psi^-_{A'A}\rangle\beta|0_B\rangle \\
&+ \frac{1}{2}|\phi^+_{A'A}\rangle\beta|1_B\rangle + \frac{1}{2}|\phi^+_{A'A}\rangle\alpha|0_B\rangle - \frac{1}{2}|\phi^-_{A'A}\rangle\beta|1_B\rangle + \frac{1}{2}|\phi^-_{A'A}\rangle\alpha|0_B\rangle \\
&= \frac{1}{2}|\psi^+_{A'A}\rangle(\alpha|1_B\rangle + \beta|0_B\rangle) \\
&+ \frac{1}{2}|\psi^-_{A'A}\rangle(\alpha|1_B\rangle - \beta|0_B\rangle) \\
&+ \frac{1}{2}|\phi^+_{A'A}\rangle(\beta|1_B\rangle + \alpha|0_B\rangle) \\
&+ \frac{1}{2}|\phi^-_{A'A}\rangle(-\beta|1_B\rangle + \alpha|0_B\rangle)
\end{aligned}$$

After doing this, we can see that

$$\begin{aligned}
|\Psi_{A'AB}\rangle = &\frac{1}{2}|\psi^+_{A'A}\rangle(\alpha|1_B\rangle + \beta|0_B\rangle) \\
&+\frac{1}{2}|\psi^-_{A'A}\rangle(\alpha|1_B\rangle - \beta|0_B\rangle) \\
&+\frac{1}{2}|\phi^+_{A'A}\rangle(\beta|1_B\rangle + \alpha|0_B\rangle) \\
&+\frac{1}{2}|\phi^-_{A'A}\rangle(-\beta|1_B\rangle + \alpha|0_B\rangle)
\end{aligned}$$

Something interesting that we can observe is that initially there was entanglement between qubits A and B, but now, there is entanglement between A' and A, and no entanglement with qubit B. According to the Born rule, if the observable $\mathcal{O}$ had 4 outcomes that correspond specifically to the four state vectors represented above ($|\psi^+_{A'A}\rangle$, $|\psi^-_{A'A}\rangle$, $|\phi^+_{A'A}\rangle$, and $|\phi^-_{A'A}\rangle$), the probability of finding

each outcome is $\frac{1}{4}$. So, when Alice measures the observable in this way, the original entanglement between qubits A and B will be swapped for the entanglement between qubits A' and A. Additionally, these 4 state vectors are still orthonormal, it's only the entangled qubits that have changed. Something that we need to note is that, since the observable has four possible outcomes, we would need exactly 2 classical bits of information to identify a specific one of its results (since a system of two qubits has four total combinations - 11, 00, 01, 10 - these combinations can be used to represent each of the four outcomes).

When Alice measures qubit A', quantum physics tells us that the state vector collapses into one of the four terms of the sum. Let us imagine that it collapses into the second term:

$$|\Psi_{A'AB}\rangle = \frac{1}{2}|\psi^{-}_{A'A}\rangle(\alpha|1_B\rangle - \beta|0_B\rangle)$$

We can observe that, when the measurement occurs, the state vector for Bob's qubit collapses too, and is defined into one of the options. Not only that, but as we can see, the complex numbers α and β that were initially set up by Alice in qubit A' have been transferred ("teleported") to qubit B without ever even getting close to it. If Alice were to send, encoded in two classical bits, which of the four Bell states her qubit collapsed into, Bob could know the state his qubit is in without needing to measure it. Therefore, with only two classical bits as information, Alice has effectively teleported information (the complex numbers) to Bob using quantum entanglement and quantum swapping. This is the phenomenon of quantum teleportation, and the mechanism through which it can teleport information from one qubit to another [3]. Notice (again) that Special Relativity is not violated here, since Bob does not receive any instantaneous information after Alice measures her qubit, and she has to send the information at speeds lower than the speed of light in the form of classical bits.

3 Applications of quantum teleportation

Any theory related to information being transported over large distances naturally lends itself to computing, and quantum teleportation is no different. It has been particularly revered because of its potential application in the fields of quantum cryptography and the quantum internet, unfortunately, in the case of quantum internet, this interest has usually been rooted in the wrong reasons. There are many articles and think-pieces that revere it as being "faster than the speed of light", which as has been previously demonstrated in this report, is simply not true. Still, there are many other reasons to be curious about the possibilities of a quantum internet and a significant amount of researchers are exploring how this technology could actually be applied.

For example, a team of researchers at Harvard in the spring of 2024 was the first to create a city-sized demo of a quantum computer network in Boston [6]. The teleportation happened in so-called memory nodes, devices, whose role it is to store individual qubits. Knaut and his team decided to use silicon-vacancy (SiV) centers in diamonds coupled to nanophotonic cavities, which is a very conveniently optically active diamond [6]. Then, these memory nodes were connected with a telecom fiber network spanning 35 km (or 27 miles), spanning Boston, Cambridge, and Sommerville [6]. In this experiment, quantum teleportation occurred when the quantum state of a photon was transferred from one memory node to another, mediated by qubits traveling through fibers [6].

Of course, for a working quantum internet in the way that is being promised by many, we would need significantly more nodes than two [6]. Furthermore, there are many drawbacks, such as the obvious extreme fragility of the quantum states, but also many promised benefits, such as better encryption, which is directly related to another emerging field in the quantum world, quantum cryptography [7].

4 Conclusions

Quantum teleportation may have begun as a result of a disagreement between Niels Bohr and Albert Einstein, but nearly 100 hundred years after the Solvay conference, it has emerged as the cornerstone of quantum physics and the foundation of many new exciting inventions. Thanks to quantum entanglement, a normal bit can suddenly be not just "off" and "on", becoming a qubit. Furthermore, through quantum swapping, the information that is encoded in one qubit can be transferred to another without any direct physical contact. While this still does not mean that things can travel faster than light, it

is still an incredibly fascinating concept with many potential uses, such as the quantum internet or quantum cryptography. Ultimately, quantum teleportation is a testament to the many strange rules of quantum physics that we have managed to understand, slowly paving the way for technologies that once seemed like science fiction.

5 Distribution of work

(Note: the project, including calculations, presentation, and solving of the problems was done in group. These are the distributions of the actual writing of the report, not the solving.)

Introduction, parts a, b, and j of project - Dafne Unsal Nuchi

Parts c to i - Abdulaziz Sobirov

Conclusions and applications (plus bibliography and making the beautiful presentation visuals and animations!) - Jolana Kampfová

References

[1] David D. Nolte. A short history of quantum entanglement. [Online; accessed 16-December-2024].

[2] A. Einstein, B. Podolsky, and N. Rosen. Can quantum-mechanical description of physical reality be considered complete? *Phys. Rev.*, 47:777–780, May 1935.

[3] Jacob Barandes, Filip Niewinski. Physics 19 office hours.

[4] John Oncea. Unveiling scientific frontiers: The impact of solvay conferences on modern science, 2023. [Online; accessed 16-December-2024].

[5] José Luis Crespo. El día que einstein casi mata a la cuántica — entrelazamiento cuántico. [Online; accessed 16-December-2024].

[6] Can M Knaut, Aziza Suleymanzade, Yan-Cheng Wei, Daniel R Assumpcao, Pieter-Jan Stas, Yan Qi Huan, Bartholomeus Machielse, Erik N Knall, Madison Sutula, Gefen Baranes, et al. Entanglement of nanophotonic quantum memory nodes in a telecommunication network. *arXiv e-prints*, pages arXiv–2310, 2023.

[7] Venus Kohli. The pros and cons of quantum networking. [Online; accessed 16-December-2024].